

Name:

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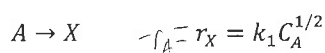
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## CHEMICAL REACTION ENGINEERING

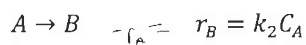
## MIDTERM EXAM 2

CHE 321, SPRING 2026, CLOSED BOOK, TIME: 75 MIN

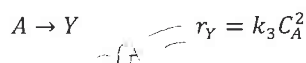
Consider the following liquid phase reactions:



$$k_1 = 0.004 \left( \frac{\text{mol}}{\text{dm}^3} \right)^{1/2} \cdot \text{min}^{-1}$$



$$k_2 = 0.3 \text{ min}^{-1}$$



$$k_3 = 0.25 \frac{\text{dm}^3}{\text{mol}} \cdot \text{min}^{-1}$$

$$\frac{r_B}{r_X} = \frac{3 C_A}{(0.004 C_A^{1/2}) + (0.25 C_A^2)^{1/2}}$$

$$= \frac{3 C_A}{0.004 + 0.5 C_A}$$

$$= 0.6$$

The reactions are carried out isothermally first in a continuously stirred tank reactor (CSTR) to maximize the overall selectivity of the desired product B over the undesired ones X and Y. The outlet of the CSTR is then fed to an isothermal plug flow reactor (PFR) to increase the conversion of species A to 99%. Species A is fed to the first reactor at a concentration of  $0.2 \frac{\text{mol}}{\text{dm}^3}$  in a liquid stream with a volumetric flow rate of  $10 \frac{\text{dm}^3}{\text{min}}$ .

1- Calculate the volume of the CSTR. What is the overall selectivity and conversion in the CSTR?

2- Calculate the volume of the PFR.

$$F_{A0} = C_{A0} \cdot v_0$$

Note: for part 2, if your calculator does not evaluate definite integrals, leave your answer in the form of a definite integral.

$$r_1 = r_3 = k_1 \cdot k_3 \cdot C_A$$

$$r_1 = r_3 = r_{\text{un}} = 0.001 C_A$$

$$r_2 = 0.3 C_A$$

$$\frac{r_2}{r_{\text{un}}} = \frac{0.3}{0.001} \frac{C_A}{C_A}$$

$$\frac{300 \frac{\text{mol}}{\text{mol}}}{10 \frac{\text{mol}}{\text{mol}}} = \frac{r_B}{r_X + r_Y}$$

$$X = \frac{F_{A0} - F_A}{F_{A0}}$$

$$r_1 + r_2 + r_3 = r_{\text{tot}} = k_1 + k_2 + k_3 \cdot C_A^2$$

$$r_{\text{tot}} = 3 \cdot 10^{-4} \cdot C_A^2$$

$$r_{\text{tot}} = 3 \cdot 10^{-4} \cdot \left( \frac{C_A}{V} \right)^2$$

$$-\frac{1}{r_A} = -3 \cdot 10^4 \cdot \frac{1}{C_A^2}$$

$$C_A = \frac{N_A}{V}$$

$$C_{A0} = 0.2$$

$$C_A = C_{A0} - C_{A0} X$$

$$C_A = 0.2 - 0.2 X$$

$$V = X \cdot \frac{F_{A0}}{r_A}$$

$$V = 0.9 \cdot \frac{10}{3 \cdot 10^{-4} \cdot (0.2 - 0.2 \cdot 0.9)^2}$$

$$V = 0.9 \cdot \frac{10}{3 \cdot 10^{-4} \cdot (0.02)^2}$$

$$V = 38.4 \text{ dm}^3$$

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$$1) \quad r_X \cdot r_Y = K_1 \cdot K_3 \cdot C_A^{\frac{1}{2}} \cdot C_A^2 \\ = K_1 \cdot K_3 \cdot C_A^{2.5}$$

$$\frac{r_B}{r_{XY}} = \frac{K_2 C_A}{K_1 \cdot K_3 C_A^{2.5}} = \frac{K_2}{K_1 \cdot K_3} \cdot C_A^{-1.5}$$

$$\boxed{\frac{r_B}{r_{XY}} = 300 C_A^{-1.5}}$$

↓  
Lower  $C_A$  at exit better selectivity

$$r_{tot} = r_X + r_B + r_Y$$

$$.0004 \cdot C_A^{\frac{1}{2}} + .3 \cdot C_A + .25 \cdot C_A^2$$

$$V_f = 2 \int_{.8}^{.99} .0004 \cdot C_A^{\frac{1}{2}} dA + \int_{.8}^{.99} .3 \cdot C_A dA + \int_{.8}^{.99} .25 \cdot C_A^2 dA$$

$$C_A = C_{A0} - C_{A0} \tau$$